

Evaluating the Fetal and Infant Safety of Doxycycline

Protocol Version: 0.1

Version Date: 08-27-2026

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Acknowledgement

The analysis is based in part on work from the Observational Health Data Sciences and Informatics collaborative. OHDSI (<http://ohdsi.org>) is a multi-stakeholder, interdisciplinary collaborative to create open-source solutions that bring out the value of observational health data through large-scale analytics.

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Rationale and Background

Doxycycline is a tetracycline class broad-spectrum antibiotic effective against several serious infections (e.g., *Chlamydia trachomatis*, malaria, Lyme disease, Rocky Mountain Spotted Fever) for which timely treatment prevents severe morbidity. Historically, tetracyclines are not prescribed to pregnant women due to fear of teratogenic risks, including major congenital malformations (MCMs), even when it represents the most effective therapy, leading to use of inferior regimens for chlamydia, limitations in syphilis treatment options, and exclusion from emerging prevention strategies such as doxycycline post-exposure prophylaxis (doxy-PEP).¹⁻⁷

Randomized controlled trials evaluating doxycycline use during pregnancy are infeasible due to ethical concerns. Large-scale observational studies have been limited, though a recent population-based cohort study of 69,656 infants in Sweden found no significant increase in risk of MCMs among 6,340 tetracycline-exposed infants (approximately 78% to doxycycline) in the first trimester (relative risk 1.03; 95% CI: 0.90–1.16).⁸ The study authors acknowledged power limitations for several MCM subgroups and individual malformations. Further, systematic review and meta-analyses illustrate that the underlying infection and its correlates (e.g., fever) also raise risks of congenital anomalies and adverse outcomes that may lead to confounding by indication.⁹

Resolving the safety profile of doxycycline during pregnancy is critical for improving maternal and child health globally. Reliable, generalizable evidence on doxycycline safety will have immediate clinical and public health implications. To date, no global, multi-database assessment has systematically evaluated doxycycline's teratogenic safety across trimesters, indications, and doses while accounting for residual confounding and outcome- or database-specific heterogeneity. The goal of this study is to leverage the Observational Health Data Science and Informatics (OHDSI) collaborative, the world's largest federated network of standardized electronic health record and claims data, to perform the first global, empirically calibrated assessment of doxycycline's fetal and infant safety.¹⁰

Study Objectives

The objective of this study is to examine the association, if any, between prenatal exposure to doxycycline and increased risk of major congenital malformations (MCMs) compared with pregnancy-appropriate alternative antibiotics prescribed for the same infection. The specific aims are:

Aim 1: To estimate unadjusted and adjusted risks of any MCM diagnosed within the first year of life, overall and by data source, using empirical calibration and extensive diagnostics.

Hypothesis: Prenatal exposure to doxycycline is not associated with an increased risk of overall MCMs compared with active, pregnancy-appropriate antibiotic comparators treating the same indication.

Aim 2: To perform subgroup and sensitivity analyses by trimester, exposure intensity, and indication.

Hypothesis: Associations (if any) between doxycycline exposure and fetal/infant outcomes vary by timing (trimester), cumulative dose/duration, and clinical indication.

Research methods

Study Design

Using a multinational retrospective active-comparator, new-user (ACNU) cohort design,¹¹ we will evaluate trimester-specific, indication-specific, and cumulative-dose risks of fetal and infant outcomes. The ACNU cohort design compares new initiators of doxycycline to new initiators of active pregnancy-appropriate comparator antibiotics for the same indication (e.g., azithromycin or amoxicillin for chlamydia; amoxicillin or cefuroxime for Lyme disease). Aligning exposure timing and indication minimizes confounding by indication, a key source of bias.

Study Population

The project will be executed across partners in the OHDSI federated network, which harmonizes electronic health record and claims data to the Observational Medical Outcomes Partnership (OMOP) Common Data Model.¹² Analyses will employ the Large-scale Evidence Generation and Evaluation across a Network of Databases (LEGEND) open-science analytics framework, which enables large-scale, reproducible, and empirically calibrated observational research.¹³ OHDSI spans 54 countries encompassing more than 950 million unique patient records and has developed OMOP-compatible pregnancy episode and mother–infant linkage algorithms that support inference of gestational timing and outcomes.^{14–15} Cohort definitions and concept sets will be developed using the OHDSI ATLAS tool. Data will remain locally controlled; partner sites will execute a common analytic package and share only aggregated results, which will be compiled by the study team.

Inclusion Criteria

(a) Pregnant individuals with ≥ 1 year pre-pregnancy observation and known pregnancy outcome, with linked infant records and ≥ 12 months infant follow-up (extending to 36 months where available to capture late-diagnosed anomalies)

(b) ≥ 1 prescription or dispensing record for doxycycline or comparator antibiotic during pregnancy. Thus, the analytic cohort will consist of mother–infant dyads, allowing evaluation of fetal and infant outcomes from birth through early childhood (0–3 years).

Exclusion Criteria

Exposure to known teratogens (e.g., isotretinoin, valproate), incomplete gestational timing, or missing linkage data.

Control Group

Active comparators are pregnancies treated with pregnancy-appropriate alternative antibiotics for the same infection type, yielding a clinically relevant counterfactual and reducing bias from infection severity or comorbidity.

Exposure

Defined as ≥ 1 prescription or dispensing of doxycycline during pregnancy. Attributes include trimester, duration, and cumulative dose. Trimester-specific sub-cohorts will enable evaluation of developmental risk periods.

Outcomes

Primary Outcomes

Any MCM diagnosed within 12 months of birth, identified through standardized OMOP concept sets.

Secondary Outcomes

Organ-specific anomalies (cardiac, neural, ocular, skeletal), stillbirth, preterm birth (<37 weeks), small-for-gestational-age, and perinatal mortality.

Exploratory Outcomes

Dental enamel or skeletal abnormalities in early childhood, where data permit.

Data Sources

The analyses will be performed across a network of observational healthcare databases that have been transformed into OMOP CDM, version 5.4. The following databases will be included in this analysis:

- Columbia University Irving Medical Center (CUIMC) EHR
- Merative Multi-state Medicaid (MDCD)
- Merative Commercial Claims and Encounters (CCAE)
- Optum® de-identified Clinformatics Data Mart Database (Clinformatics)

Data Analysis Plan

Meta-Analysis

Site-specific estimates will be pooled via random-effects meta-analysis to derive global summary effects and quantify heterogeneity (I^2). Between-database variation will be explored via stratification and/or meta-regression by geography, data model, and health system-context.

Outcome Modeling

Cox proportional-hazards models (and, where appropriate, conditional logistic models for matched analyses) will yield adjusted hazard ratios (and/or odds ratios) with 95% CIs.

Empirical calibration

Negative controls (and synthetic positive controls) will be used to quantify residual systematic error and calibrate statistical inference (calibrated CIs and p-values).¹⁶

Diagnostics and Sensitivity Analyses

Covariate-balance plots, overlap/positivity diagnostics, and negative-control diagnostics will assess residual bias. Sensitivity analyses will examine (a) alternative exposure definitions, (b) trimester-specific exposures, (c) indication-restricted subsets, (d) high-dimensional propensity score (PS) variants, and (e) infant follow-up extended to 36 months.

Control of Confounding

The ACNU design reduces confounding by indication; remaining measured confounders will be addressed via large-scale PS modeling using regularized regression (e.g., LASSO) over predefined covariates (demographics, comorbidities, medications, procedures, infection proxies, care setting).¹⁷ We will implement 1:1 PS matching or PS-based weighting to achieve covariate balance (standardized mean difference < 0.1).

Study Diagnostics

Sample Size and Study Power

OHDSI's federated network covers approximately 974 million unique patients; statistical power will be evaluated per outcome and database.

Protection of Human Subjects

The study is collecting only de-identified data. Only aggregated results will be shared across the community. To protect patient privacy and reduce the risk of re-identification, each site will exclude data for any concept where the record count ≤ 10 , and the true counts will be randomized by the Poisson distribution. Confidentiality of patient records will be maintained at all times.

Plans for Disseminating and Communicating Study Results

The aggregated study results will be posted on the OHDSI website after completion of the study. At least one paper will be written and submitted for publication to a peer-reviewed scientific journal.

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